

Umang Singh

Backend software engineer, **2+ years** at **Oracle Financial Services**, building APIs, microservices and distributed systems in Java.

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WORK EXPERIENCE

Oracle (OFSS) – Machine Learning for Anti-Money Laundering Team (ML4AML)

Mumbai, India

Application Software Engineer

Jul 2024 – Present

- **Core developer** on the Automated Scenario Calibration (ASC) SaaS platform for anti-money laundering, delivering Java and Python services across **2** Financial Services products on OCI, and migrating its metering service from **Spring Boot to Helidon MicroProfile**.
- Redesigned provisioning error handling across **6** Java integrations between ASC SaaS and its data-science platform, separating non-retryable failures from transient ones to eliminate up to **5 minutes** of fixed waiting per failure, while retaining **bounded exponential backoff**, status polling, idempotent duplicate handling, terminal-state propagation, and health checks.
- Implemented, SQL-registered and unit-tested **24 Python** behavioral red-flag feature transformers, and integrated below-the-line alerts into the investigator case-management system across **4** data pipelines and a **15-task** orchestration workflow.
- Diagnosed a case-generation failure caused by incompatible **SQL bind metadata** and drove a targeted recovery, closing validation with all **6** sampled alerts converted to distinct cases and the negative control correctly excluded.
- Re-architected a **30,000+** line Python monolith (**200+ APIs**) into independently deployable microservices on **Docker, Kubernetes, and OCI**, decoupling ML logic from the application layer, improving testability, and driving component reuse **across 5 projects**.
- Own release engineering for the ML4AML services, covering **Liquibase** schema migrations, **Jenkins/OCI CI/CD**, and dependency remediation; built a bulk threshold-management workflow and maintain the Compliance Studio notebooks used for calibration.

Project Intern

Jun 2023 – Jul 2023

- Constructed **25+ REST APIs** and containerized **Helidon** services with Docker for ASC SaaS, and established initial deployment pipelines, condensing **50+** paragraphs of analytical notebooks to streamline the model-calibration workflow for data scientists.

Futurepreneur Tech (StayWise)

Mumbai, India

Tech Lead

Feb 2023 – May 2023

- Architected a scalable **Firebase** real-time backend and shipped the mobile application in **90 days**, onboarding **20+ hostels** across Mumbai while managing a **15-member** team across UI/UX, frontend, and backend and securing **INR 100,000** in seed funding.

EDUCATION

University of Mumbai | [TSEC Hall of Fame](#) (30 of 600+)

Mumbai, India

Bachelor of Engineering, Computer Engineering | **GPA: 8.76/10**

2020 – 2024

PROJECTS

Krishibodha: Agentic AI Solution for Farmers | GCP, Gemini, Gemma, Vertex AI, Google ADK, Flutter

Jun 2025 – Aug 2025

- Designed the agent orchestration layer in **Google ADK**, using Gemini to reason over a crop, market, and subsidy knowledge base and generate crop recommendations, with an on-device **Gemma** fallback that keeps the assistant answering when connectivity drops.
- Served the models on **Vertex AI** and unified multilingual voice input, image-based crop-disease diagnosis, localized market insights and alerts, and satellite crop-and-soil-health monitoring behind a single **Flutter** client usable offline.

VidyuRakshak: Satellite-Imagery-Based Transmission-Line Safety | Google Earth Engine, ML, Flutter, AWS

Dec 2023 – Aug 2024

- Developed end-to-end geospatial ML pipelines for power-corridor risk analysis using Landsat and stereo imagery, training a **Random Forest Regressor** across **7 vegetation indices** and validating canopy heights against a **107,000-tuple** DTM/DSM dataset.
- [Selected by Power Grid Corporation of India for continued development under a signed agreement](#), funding a second proof of concept on **AWS** (S3, Lambda, API Gateway) for automated tree-growth monitoring and predictive analytics.

PUBLICATIONS

- [“Advanced Deep Learning Techniques for Early Detection of Oral Cancer: A Multimodal Approach”](#); ISAI 2025, Springer. Mar 2025
- [“Exploring Challenges and Results in Crop Type Mapping Using Satellite Imagery”](#); IEEE InGARSS 2024. **First author**. Dec 2024
- [“Tree-height estimation along power-line corridors from optical and stereo satellite imagery”](#); ICICBDA 2024, Springer. Jun 2024

AWARDS AND ACHIEVEMENTS

- Led a team to **1st Runner-Up** at [Google Agentic AI Hackathon 2025](#), a **Guinness World Record** event with **57,000+** registrants.
- **1st Runner-Up** at [Japan-India Hackathon 2024](#), organized by IIT Gandhinagar and NEDO Japan, with a prize of **INR 150,000**.
- Winner of [Smart India Hackathon 2023](#), a Government of India initiative and India's largest hackathon, with **250,000+** participants.
- **Top 8 of 11,000+** at the BIAL Genie Hackathon, powered by Kempegowda International Airport and Microsoft.
- Ranked **1,081 of 24,578 (top 4.4%)** in LeetCode [Weekly Contest 352](#), with **690+** problems solved across **500+** active days.
- **Mentored 15** teams on the AI/ML track at [HackSync 2.0](#) at the invitation of GDG TSEC; taught mathematics at U&I NGO for two years.

SKILLS

Languages & Backend: Java, Python, SQL, Kotlin, Helidon MicroProfile, Spring Boot, REST APIs, microservices, Flutter

Cloud & DevOps: OCI, GCP, AWS (Lambda, S3), Firebase, Docker, Kubernetes, Jenkins, Liquibase, Terraform, CI/CD, Git, Bash

AI/ML: LLMs, RAG, agentic workflows (Google ADK), Gemini, Gemma, Vertex AI, PyTorch, Hugging Face

AI-assisted development: Claude Code, Codex, GitHub Copilot; prompt engineering, review and validation of generated code